

Gunter & Scott 1990 — Inventory of Definitions and Theorems

Source: **C. A. Gunter and D. S. Scott**, “**Semantic Domains**,” *Handbook of Theoretical Computer Science* Vol. B, North-Holland, 1990, pp. 633–674 ([../papers/Gunter Scott 1990.pdf](#)).

The work list for the Lean formalization: every definition and every one of the paper’s **30 numbered results** (Theorems / Lemmas / Proposition 1–30), in paper order, matched to its Lean equivalent.

Progress (as of r0024, 2026-0806)

#	Quantity	Done	Remaining	Of
1	Definitions to define	8	5	≈13
2	Numbered results complete	7 (Thm 1, Thm 3, Lem 4, Lem 5, Thm 6, Thm 7, Lem 8)	22	29
3	Unnumbered prose claims proved	12	—	—
4	Mathlib foundations reused	12	—	12
5	Theorems in the development	133 live (+6 commented out as unused)	—	—
6	sorry in the development	—	0	—

Row 2 counts only the paper’s 30 **numbered** results (Theorems / Lemmas / Proposition 1–30). Row 3 counts the claims the paper makes **in prose** rather than as numbered results; these are paper content too, and all twelve are formally verified:

#	Paper claim	Where	Lean
1	“the compact elements [of $P \rightarrow N$] are just the finite subsets of N ”	p. 9	<code>isCompactElement_iff_finite</code>
2	“ $P \rightarrow N$... is a domain”	p. 9	<code>instance : Domain (Set X)</code>
3	“ $D \rightarrow E$ is a ... cpo”	Thm 7 proof	<code>instance : CompletePartialOrder (ScottHom α β)</code>
4	“... bounded complete ... whenever E is”	Thm 7 proof	<code>instance : BoundedComplete (ScottHom α β)</code>
5	“step(s) ... is continuous”	Thm 7 proof	<code>scottContinuous_stepFun</code>
6	“... and compact in the ordering on $D \rightarrow E$ ”	Thm 7 proof	<code>isCompactElement_step</code>
7	“they form a basis for $D \rightarrow E$ ”	Thm 7 proof	<code>instance : IsAlgebraic (ScottHom α β)</code>
8	every compact function is a <i>finite</i> join of step functions	Thm 7 proof, implicit	<code>exists_finite_isLUB_of_isCo</code>
9	“an embedding is an injection”	§3.1	<code>IsEmbeddingProjectionPair.i</code>
10	“a projection is a surjection”	§3.1	<code>IsEmbeddingProjectionPair.s</code>
11	“it is easy to check that p_N ... is a finitary projection”	§3.1	<code>isFinitaryProjection_normal</code>
12	“the set of strict continuous functions $D \rightarrow E$ is also a cpo”	§2.1	<code>ScottDomains.strictHomCpo</code>

Six of those eleven are the body of **Theorem 7**, which is now **complete** — all four conjuncts of its conclusion (cpo, bounded complete, algebraic, countably based) are formally verified, as `ScottHom.isBoundedCompleteDomain_scottHom`.

It is proved under **weaker hypotheses than the paper states**: bounded completeness of D is never used. D need only be a domain. The function space is a cpo for any preordered D , algebraic when D and E are, bounded complete because E is, and countably based because D and E are.

The remaining theorems are supporting API: the $\ll$ calculus, the `compactsBelow` machinery, the pointwise order and suprema on $D \rightarrow E$, and the step-function adjunction. The paper assumes or elides all of it.

Reference audit (r0020). Of the theorems in the development, 16 are never cited elsewhere. Nine of those are *terminal by design* — they are the paper’s own claims, so nothing should cite them (`injective_embedding`, `surjective_projection`, the `isNormalIn_sUnion*` family and `mono_right` for Lemma 4, `singleton_bot_isNormalIn` for $\{\perp\} \in P(C)$, `isLeast_kleeneFix_le`, `eq_kleeneOperator_op`). The other six were speculative API written for callers that never appeared; they are **commented out in place**, each with a note on why it exists and what is instructive about it, and the build is unchanged — which also confirms that the three `@[simp]` ones among them were never firing implicitly.

The development is **21 modules, 3033 lines, 0 sorry, 0 warnings**. Counts of definitions, results and theorems are in the Progress table above — they are not repeated here, so that this section cannot drift out of step with it. What each round delivered:

#	Round	Module	Contents
1	r0003	<code>ScottDomains/WayBelow.lean</code>	<code>way-below</code> $\ll$ at $[Preorder\ \alpha]$, 7 theorems; $x \ll x \leftrightarrow IsCompactElement\ x$ holds by <code>Iff.rfl</code>
2	r0004	<code>ScottDomains/Domain.lean</code>	<code>IsAlgebraic</code> , <code>Domain</code> (with the paper’s countable-basis condition), <code>BoundedComplete</code> , 8 theorems, <code>Domain Prop</code>
3	r0005	<code>ScottDomains/Powerset.lean</code>	the paper’s $P\ \mathbb{N}$ (p. 9): compacts of <code>Set X</code> are exactly the finite subsets, hence <code>Domain (Set ℕ)</code> — the nondegenerate witness
4	r0006–r0007	<code>ScottDomains/ScottHom.lean</code>	the continuous function space $D \rightarrow E$: <code>ScottHom</code> , the pointwise order, <code>CompletePartialOrder</code> , and <code>BoundedComplete</code> when E is — Theorem 7’s first sentence in full
5	r0008	<code>ScottDomains/StepFunction.lean</code>	the single step function <code>step k e</code> : continuity (from <code>k compact</code>), the adjunction <code>step k e ≤ f ↔ e ≤ f k</code> , and compactness in $D \rightarrow E$ (from <code>e compact</code>)
6	r0009	<code>ScottDomains/FunctionSpaceDomain.lean</code>	<code>IsAlgebraic</code> — the paper’s “they form a basis for $D \rightarrow E$ ”; the two halves of <code>IsAlgebraic</code> use disjoint hypotheses (directedness needs only E bounded complete; the lub needs only D, E algebraic)
7	r0010	<code>ScottDomains/CompactFunctionSpace.lean</code>	every compact function is a finite join of step functions — the finiteness half of the basis claim
8	r0011	<code>ScottDomains/FunctionSpaceCountable.lean</code>	<code>Countable</code> — E is countable, hence <code>Domain (ScottHom α β)</code> — Theorem 7 complete
9	r0012	<code>ScottDomains/NormalSubposet.lean</code>	the normal-subposet relation $\triangleleft$ and Lemma 4 , all four parts
10	r0012–r0013	<code>ScottDomains/Projection.lean</code>	embedding–projection pairs, projections, the paper’s “an embedding is an injection, a projection is a surjection”; <code>im(p)</code> is a cpo and finitary projections

#	Round	Module	Contents
11	r0014	ScottDomains/FinitaryProject	Lemma 5 — the compacts of $\text{im}(p)$ are $\text{im}(p) \cap K(D)$ (needs only that p is a projection), and $\text{im}(p) \cap K(D) \triangleleft K(D)$
12	r0015	ScottDomains/NormalProjection	Lemma 6 — $p_N = \bigsqcup \{y \in N \mid y \sqsubseteq x\}$: continuous, a projection, and $\text{im}(p_N) \cap K(D) = N$ — half of Theorem 6 's correspondence, plus order preservation both ways
13	r0016	ScottDomains/Theorem6.lean	Theorem 6 — p_N is finitary, $p_{\{\text{im}(p) \cap K(D)\}} = p$, and the correspondence assembled
14	r0017	ScottDomains/FixedPoint.lean	Theorem 1 — $\bigsqcup_n f^n(\perp)$ is the least fixed point of a continuous f on a cpo. Not Mathlib reuse; see the §2 table
15	r0018	ScottDomains/UniformFixedPoint	Theorem 3 — fix is the unique uniform fixed-point operator; $\downarrow a$ as a cpo
16	r0019	ScottDomains/Product.lean	$D \times E$ as a cpo (the one construction needing no case split) and Lemma 8 parts 1–3
17	r0021	ScottDomains/Currying.lean	Lemma 8.4 — currying, $D \rightarrow (E \rightarrow F) \cong (D \times E) \rightarrow F$, and with it Lemma 8 complete
18	r0022	ScottDomains/EffectivePresentation	§2 is effective presentation — the enumeration of the basis with its two decidability conditions
19	r0023	ScottDomains/Lift.lean	the lift $D \downarrow$ as a cpo, on Mathlib's <code>WithBot</code>
20	r0024	ScottDomains/StrictHom.lean	the strict function space $D \rightarrow \downarrow E$ as a cpo — needs no case split, since both branches of <code>ScottHom</code> 's <code>sSup</code> are strict

§2 and §3 are now complete — the only §3 omission is the paper's *computable function*, which needs an r.e.-predicate notion Mathlib does not supply.

Dependency structure of the five remaining definitions

#	Definition	Prerequisites	Status
1	smash product $D \otimes E$	Domain, OrderBot — both built	independent, ready
2	sum $D + E$	Domain — built	independent, ready
3	bifinite / Plotkin order	IsNormalIn ($\triangleleft$, r0012) — built	independent, ready
4	D_∞	embedding–projection pairs (r0012) — built	independent, but large (inverse limits)
5	the three powerdomains	the ideal completion (Theorem 11) — <i>not</i> built	blocked ; the three are independent of each other once it exists

Rows 1–3 are mutually independent with every prerequisite in place, so they can be written as three separate modules and checked in a single build.

Next: those three, then Theorem 11 to unblock the powerdomains, then Lemma 9 (the strict analogues of Lemma 8) and Lemma 10 (closure of bounded completeness).

Work counts

- **Reuse from Mathlib — no work (11):** poset, directed, cpo, $\perp$, monotone, continuous, fixed-point operator, Schröder–Bernstein (2), algebraic lattice, product, λ -notation. (Theorem 1 was listed here in an earlier draft and has been removed: Mathlib’s `OrderHom.lfp` is Knaster–Tarski over a complete lattice, not Kleene’s $\bigsqcup_n f^n(\perp)$ over a cpo. It is proved in `ScottDomains/FixedPoint.lean`, so **29** numbered results needed proof, not 28.)
- **Generalize / adapt (4):** compact element (`IsCompactElement` — its *definition* is already stated at `[PartialOrder  $\alpha$ ]` and so applies to a dcpo verbatim; what is `CompleteLattice`-only is every lemma about it, so the dcpo API must be re-proved), sum, lift (`WithBot/Sum`), ideal completion (`Order.Ideal`).
- **Definitions to define — new (≈ 13 ; 4 done in r0003–r0004, 9 remaining):** way-below $\ll$ (**done** — `ScottDomains/WayBelow.lean`), algebraic cpo, **domain**, bounded-complete (**all three done** — `ScottDomains/Domain.lean`), embedding–projection pair, (finitary) projection, normal subposet, effective presentation, smash product, the three powerdomains (Hoare / Smyth / Plotkin), bifinite / Plotkin order, and D_∞ .
- **Theorems to prove (28):** 28 of the paper’s 30 numbered results — Theorems 3, 6, 7, 11, 12, 14, 16, 18, 21, 22, 25, 26, 27, 29; Lemmas 4, 5, 8–10, 13, 17, 19, 20, 23, 24, 28, 30; Proposition 15. Only Theorems 1 & 2 come free from Mathlib.

Bottom line: ≈ 13 definitions to define + 28 results to prove, on top of 12 reused Mathlib foundations. After r0004: **9 definitions + 28 results** remain.

Lean column legend: $\checkmark$ reuse Mathlib (name given) $\cdot$ ~ partial (Mathlib has a related or lattice-only version — generalize) $\cdot$ $\times$ define / prove (absent from Mathlib v4.32.2, confirmed by grep).

Statements are paraphrased/de-garbled from the PDF (1990 Type-3 fonts render $\rightarrow$ as $!$, drop fi ligatures, mangle math), so read them as a guide to *what* each result is. Notation: $\sqsubseteq$ order, $\bigsqcup$ directed sup, $\perp$ bottom, $\ll$ way-below, $K(D)$ compacts, $Fp(D)$ finitary projections.

§2 Recursive definitions of functions

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
2.1	Def	—	poset (partially ordered set)	$\checkmark$ <code>PartialOrder</code>
2.1	Def	—	directed subset M : every finite $u \subseteq M$ has an upper bound in M	$\checkmark$ <code>DirectedOn / IsDirected</code>
2.1	Def	—	cpo : poset in which every directed M has a lub $\bigsqcup M$	$\checkmark$ <code>CompletePartialOrder</code> (chains: <code>OmegaCompletePartialOrder</code>)
2.1	Def	—	bottom $\perp$ (least element)	$\checkmark$ <code>OrderBot</code>
2.1	Def	—	monotone function	$\checkmark$ <code>Monotone / OrderHom</code>
2.1	Def	—	continuous : monotone, $f(\bigsqcup M) = \bigsqcup f(M)$ for directed M	$\checkmark$ <code>ScottContinuous</code>
2.1	Thm	1	Fixed-Point Theorem : $f : D \rightarrow D$ continuous $\implies$ least fixed point $\bigsqcup_n f^n(\perp)$	$\checkmark$ proved (r0017) — <code>ScottDomains.theorem1</code> . Not Mathlib reuse: <code>OrderHom.lfp</code> is Knaster–Tarski over a <i>complete lattice</i> with only monotonicity, a different theorem
2.2	Thm	2	Schröder–Bernstein for sets	$\checkmark$ <code>Function.schroeder_bernstein</code> (<code>SetTheory/Cardinal/SchroederBernstein.lean</code> — the name in an earlier draft of this row, <code>Function.Embedding.schroederBernstein</code> , does not exist)
2.3	Def	—	fixed-point operator (uniform)	$\checkmark$ <code>OrderHom.lfp / LawfulFix</code>
2.3	Thm	3	The standard operator is the unique uniform fixed-point operator	$\checkmark$ proved (r0018) — <code>ScottDomains.theorem3</code> ; a fixed point operator is formalized as a family over every <code>CompletePartialOrder</code> in a universe

§3 Effectively presented domains

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
3.1	Def	—	compact element x : $x \sqsubseteq \bigsqcup M(\text{dir.}) \implies x \sqsubseteq y$ some $y \in M; K(D)$	<code>~ IsCompactElement</code> — def is <code>[PartialOrder]</code> , so reusable on a <code>dcpo</code> ; its lemmas are all <code>CompleteLattice</code> -only
3.1	Def	—	way-below $\ll$ (approximation relation behind compactness)	✓ <code>ScottDomains.WayBelow</code> (r0003) — $x \ll x \leftrightarrow \text{IsCompactElement } x$ by <code>Iff.rfl</code>
3.1	Def	—	algebraic cpo: $x = \bigsqcup \{x' \in K(D) : x' \sqsubseteq x\}$ (directed)	✓ <code>ScottDomains.IsAlgebraic</code> (r0004)
3.1	Def	—	domain = algebraic cpo whose basis $K(D)$ is countable (the paper’s definition, p. 9 — the countability condition was missing from an earlier draft of this row)	✓ <code>ScottDomains.Domain</code> (r0004)
3.1	Def	—	bounded complete : $\perp +$ every bounded subset has a sup — a <i>separate</i> predicate; the paper composes them as “bounded complete domain” (Thm 7, Lem 10, Lem 13, Thm 14), which is the literature’s <i>Scott domain</i>	✓ <code>ScottDomains.BoundedComplete</code> (r0004); the compound is <code>[Domain α] [BoundedComplete α]</code>
3.1	Def	—	(countably based) algebraic lattice	✓ <code>IsCompactlyGenerated</code> (+ <code>CompleteLattice</code>)
3.1	Def	—	embedding–projection pair (g, f)	✓ <code>ScottHom.IsEmbeddingProjectionPair</code> (r0012)
3.1	Def	—	projection; finitary projection p : $p \circ p = p \sqsubseteq \text{id}$, $\text{im}(p)$ a domain	✓ <code>ScottHom.IsProjection</code> (r0012), <code>ScottHom.IsFinitaryProjection</code> (r0013) — $\text{im}(p)$ carries a <code>CompletePartialOrder</code> via <code>IsProjection.rangeCompletePartialOrder</code>
3.1	Def	—	normal subposet / substructure	✓ <code>ScottDomains.IsNormalIn</code> , notation $\triangleleft$ (r0012)
3.1	Lem	4	$\langle P(C), \triangleleft \rangle$ of substructures is a cpo with $\{\perp\}$ least	✓ proved (r0012), all four parts
3.1	Lem	5	p finitary projection $\implies$ compacts of $\text{im}(p)$ are $\text{im}(p) \cap K(D)$, and $\text{im}(p) \cap K(D) \triangleleft K(D)$	✓ proved (r0014) — <code>IsProjection.isCompactElement_iff</code> (needs only <i>projection</i>) and <code>IsFinitaryProjection.isNormalIn_compacts</code>
3.1	Thm	6	Isomorphism: normal substructures $\cong \text{Fp}(D)$ (finitary projections)	✓ proved (r0015–r0016) — <code>ScottDomains.theorem6</code>
3.1	Thm	7	D, E bounded-complete domains $\implies D \rightarrow E$ bounded-complete domain	✓ proved (r0006–r0011) — <code>ScottHom.isBoundedCompleteDomain_scottHom</code> ; D bounded complete is not needed
3.2	Def	—	effective presentation $d : \mathbb{N} \rightarrow K(D)$; effectively presented domain	✓ <code>ScottDomains.EffectivePresentation</code> (r0022). The paper’s computable function is <i>not</i> formalized: it needs “recursively enumerable”, and Mathlib v4.32.2 has no <code>RePred</code> or equivalent (<code>grep</code> finds none) — building it is recursion theory, not domain theory

§4 Operators and functions

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
4.1	Def	—	product $D \times E$	✓ Prod order from Mathlib; the cpo instance is <code>ScottDomains.instCompletePartialOrderProd</code> (r0019) — Mathlib has <code>Prod.supSet</code> and <code>isLUB_prod</code> but no cpo instance
4.2	Def	—	Church's λ-notation (continuous abstraction)	✓ <code>OrderHom / ωCPO ContinuousHom</code>
4.3	Def	—	smash product $D \otimes E$	✗ define
4.4	Def	—	sum $D + E$; lift $D \perp$	lift ✓ <code>ScottDomains.liftCpo</code> on <code>WithBot</code> (r0023) — Mathlib gives the order, not the cpo; sum still ~
4.x	Lem	8	$D \times E \cong E \times D$; $(D \times E) \times F \cong D \times (E \times F)$; $D \rightarrow (E \times F) \cong (D \rightarrow E) \times (D \rightarrow F)$; $D \rightarrow (E \rightarrow F) \cong (D \times E) \rightarrow F$	✓ proved — <code>prodComm</code> , <code>prodAssoc</code> , <code>scottHomProd</code> (r0019); <code>scottHomCurry</code> (r0021)
4.x	Lem	9	Product/function-space iso laws over D, E, F	✗ prove
4.5	Lem	10	D, E bounded complete $\implies \rightarrow, \times, \otimes, +, () \perp$ bounded complete	✗ prove
4.5	Thm	11	Ideal completion of a countable pre-order is a domain (all domains so arise)	~ <code>Order.Ideal</code> exists $\rightarrow$ prove
4.5	Thm	12	Initiality of a continuous algebra satisfying axioms T	✗ prove
4.5	Lem	13	D bounded complete $\implies$ powerdomains $D], D[$ bounded complete	✗ prove
4.5	Thm	14	Equivalent characterizations of an (algebraic/BC) domain	✗ prove

§5 Powerdomains

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
5.1	Def	—	powerdomain (non-deterministic outcomes)	✗ define
5.2	Def	—	Hoare (lower), Smyth (upper), Plotkin (convex) powerdomains	✗ define
5.3	—	—	Universal & closure properties (see Lem 13, 28, 30)	✗ prove

§6 Bifinite (SFP) domains

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
6.1	Def	—	Plotkin order / bifinite (SFP) domain: bilimit of finite pointed posets	✗ define
6.1	Prop	15	Every bounded-complete domain is bifinite	✗ prove
6.1	Thm	16	D bifinite $\implies \text{Fp}(D)$ is an algebraic lattice	✗ prove
6.2	Lem	17	D, E bifinite $\implies \rightarrow, \times, \otimes, +, () \perp$ bifinite (incl. function space)	✗ prove
6.2	Thm	18	If D and $D \rightarrow D$ are domains, then D is bifinite	✗ prove
6.2	Lem	19	closure $r: D \rightarrow D$ ($r \circ r = r \sqcup \text{id}$) $\implies \text{im}(r)$ is a domain	✗ prove

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
6.2	Lem	20	$D \text{ domain} \implies Fc(D)$ (finitary closures) is a cpo	✗ prove

§7 Recursive definitions of domains (universal domain, D^∞)

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
7	Def	—	recursive domain equation; universal domain / D^∞ (inverse limit)	✗ define
7	Thm	21	F representable over cpo $U \implies$ a domain D with $D \cong F(D)$	✗ prove
7	Thm	22	Any countably-based algebraic lattice L : a closure $r : P(\mathbb{N}) \rightarrow L$	$\sim$ IsCompactlyGenerated $\rightarrow$ prove
7	Lem	23	The function-space operator is representable over $P(\mathbb{N})$	✗ prove
7	Lem	24	U cpo; $\times$ and $\rightarrow$ representable over $U \implies$ (setup for universality)	✗ prove
7	Thm	25	U non-trivial domain representing $\times, \rightarrow \implies$ U universal	✗ prove
7	Thm	26	Any signature $(s_1, \dots, s_n)$: combinators $F_1, \dots, F_n$ solving the equations	✗ prove
7	Thm	27	Any bounded-complete D : a projection of the universal domain onto D	✗ prove
7	Lem	28	Operators $\rightarrow, \times, \otimes, +, ()^\perp, ()], ()[$ representable over U	✗ prove
7	Thm	29	D bifinite $\implies D^+$ bifinite; solving $D \cong D^+$	✗ prove
7	Lem	30	Operators $\rightarrow, \times, \otimes, +, ()^\perp, ()], ()[$ p-representable over universal bifinite V	✗ prove

Tally (matched): ✓ reuse Mathlib $\approx$ **12** (poset, directed, cpo, $\perp$, monotone, continuous, fixed-point Thm 1 & operator, Schröder–Bernstein, algebraic lattice, product, λ -notation) · $\sim$ partial $\approx$ **4** (compact element, sum/lift, ideal completion Thm 11, Thm 22) · ✗ define / prove $\approx$ **44**, of which 4 are done ($\ll$ r0003; algebraic, domain, bounded-complete r0004) $\rightarrow$ **40 remaining** — the bifinite / powerdomain / D^∞ development and all 28 numbered results.

What is done, and what is next, is listed under **Progress** at the top of this file.